1.解：(1) $u=\frac{24}{20}=1.2$，$d=\frac{19}{20}=0.95$　$1+r=1+\frac{10\%}{4}=1.025$

则 $p_1=\frac{1.025-0.95}{1.2-0.95}=\frac{0.075}{0.25}=0.3$，$p_2=0.7$

(***) 10%是年利，单期应为$\frac{10\%}{4}$

于是状态空间 $\{w_i \mid i=1,2\}$，w_1表示股价上升，w_2表示股价下降

测度 $\mathbb{P}$：$\mathbb{P}(w_1)=0.3$，$\mathbb{P}(w_2)=0.7$

期末股票 $S \sim$

24	19
0.3	0.7

(2) $u=1.1$，$d=0.9$，$1+r=1.025$

状态空间：$\{(w_i,w_j) \mid i,j=1,2\}$，$w_1$表示股价上升，$w_2$表示股价下降

单期内上涨 $p=\frac{1+r-d}{u-d}=\frac{1.025-0.9}{1.1-0.9}=\frac{0.125}{0.2}=0.625$，$1-p=0.375$

测度 $\mathbb{Q}$：

(w_1,w_1)	(w_1,w_2)	(w_2,w_1)	(w_2,w_2)
p^2	$p(1-p)$	$(1-p)p$	$(1-p)^2$
0.390625	0.234375	0.234375	0.140625

股票在期末的价格服从：

24.2	19.8	16.2
p^2	$2(1-p)p$	$(1-p)^2$
0.390625	0.46875	0.140625

(3) $u_1=1.1$，$d_1=0.9$，$u_2=1.08$，$d_2=0.88$　$1+r=1.025$

状态空间：$\{(w_i,\theta_j) \mid i,j=1,2\}$　w_1,w_2表示在第一期涨，跌

θ_1,θ_2表示在第一期涨，跌

$p(w_1)=\frac{1.025-0.9}{1.1-0.9}=0.625$，$p(w_2)=0.375$

$p(\theta_1)=\frac{1.025-0.88}{1.08-0.88}=\frac{0.145}{0.2}=0.725$，$p(\theta_2)=0.275$

测度Q: (w_1, θ_1) (w_1, θ_2) (w_2, θ_1) (w_2, θ_2)

0.453125 0.171875 0.271875 0.103125

期末价格: 第一期:

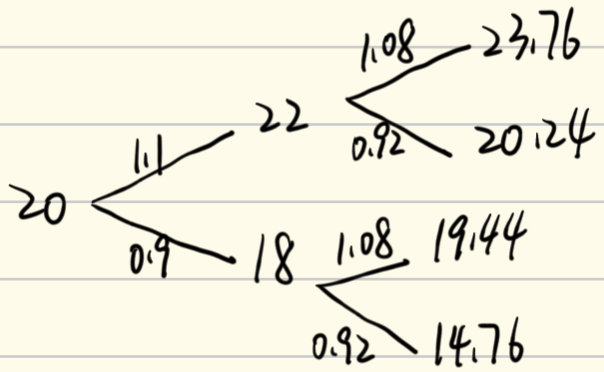
P	22	18
q_1	0.625	0.375

第二期:

P	23.76	19.36	19.44	15.84
q_2	0.453125	0.171875	0.271875	0.103125

2. 解:

(1)

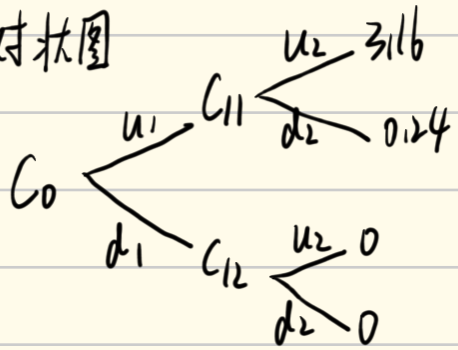


$e^{\frac{0.12}{12}\times 3} = e^{0.03} \approx 1.03$, $0.9 < 1.03 < 1.1$, $0.92 < 1.03 < 1.08$, 市场无套利

(2) $P_{11} = \dfrac{e^{0.03} - 0.9}{1.1 - 0.9} \approx \dfrac{0.13}{0.2} = 0.65$ $P_{12} = 0.35$

$P_{21} = \dfrac{e^{0.03} - 0.92}{1.08 - 0.92} = \dfrac{0.11}{0.16} = \dfrac{11}{16} = 0.6875$, $P_{22} = \dfrac{5}{16} = 0.3125$

期权树状图



其中 $C_{12} = 0$

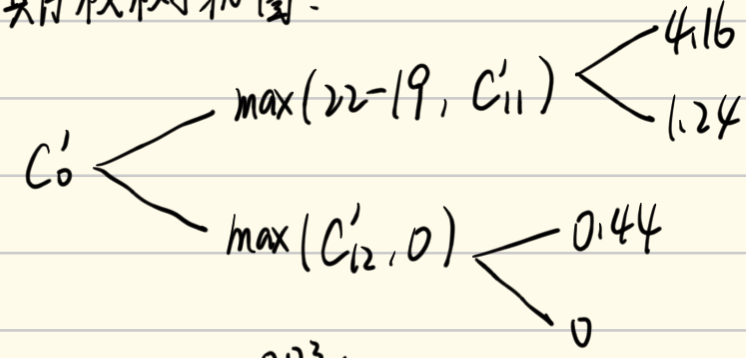
$C_{11} = e^{-0.03}(P_{21} \times 3.16 + P_{22} \times 0.24)$

$= e^{-0.03} \times 2.234825$

≈ 2.17

$C_0 = e^{-0.03}(P_{11} \times 2.17 + P_{12} \times 0) \approx 1.37$ (元)

(3) 期权树状图：

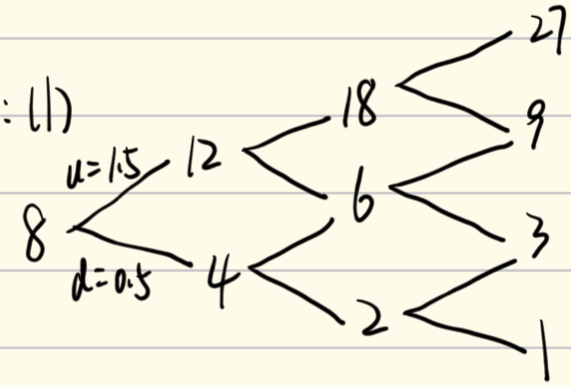


其中 $C_{11}' = e^{-0.03}(P_{11} \times 4.16 + P_{12} \times 1.24)$
≈ 3.15

$C_{12}' = e^{-0.03}(P_{11} \times 0.44)$
≈ 0.29

则 $C_0' = e^{-0.03}(3.15 \times 0.65 + 0.29 \times 0.35) \approx 2.09$ (元)

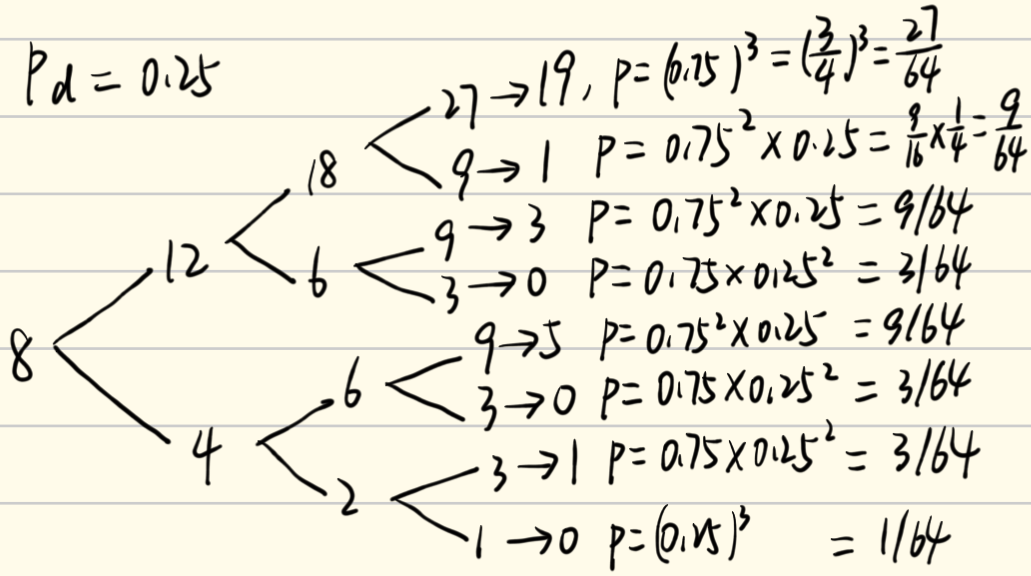
3. 解：(1)



$$P_u = \frac{1+0.25-0.5}{1.5-0.5} = 0.75$$

$P_d = 0.25$

(2)



$$C_0 = (1+0.25)^{-3}\sum\left(S_3 - \min_{123} S(w)\right)\times P = \frac{9.328125}{1.25^3} \approx 4.78$$

记得贴现